

Vignette: Basic Implementation Check (Permutation Test on ECG Age Gap)

1. Overview

This package implements the core workflow for our project:

a **two-sample permutation test** comparing participants with atrial fibrillation (AF) versus those without AF on an ECG-derived continuous outcome called *Age Gap*:

$$g = \text{nn_predicted_age} - \text{age}.$$

The goal is to test whether the distribution of g differs between AF and non-AF groups using a nonparametric permutation procedure, following the randomization-testing framework of Fisher.

2. Functions Implemented

The basic pipeline uses four key functions:

1. `load_ecg_data(path)`

Reads the dataset metadata file and keeps the needed columns:

`patient_id`, `exam_id`, `age`, `nn_predicted_age`, `AF`.

2. `filter_to_unique_patient(df, method = "latest" | "random", seed = ...)`

Keeps **one exam per patient** to reduce repeated-measures dependence.

3. `compute_age_gap(df)`

Adds the variable

$$\text{age_gap} = \text{nn_predicted_age} - \text{age}.$$

4. `run_permutation_test(x, y, stat = "mean" | "median" | "ks", B, seed)`

Runs the permutation test and returns `t_obs`, `t_perm`, `p_value`, and related metadata.

3. Demonstration with Synthetic Data

For the vignette, we use a small synthetic dataset so that anyone can run the example without needing the real CODE-15% files.

```

set.seed(1)

n_patients <- 200
exams_per_patient <- sample(1:3, n_patients, replace = TRUE)
N <- sum(exams_per_patient)

df <- data.frame(
  patient_id = rep(seq_len(n_patients), times = exams_per_patient),
  exam_id = sample(100000:500000, N, replace = FALSE),
  age = sample(18:95, N, replace = TRUE),
  nn_predicted_age = round(rnorm(N, mean = 55, sd = 12), 5),
  AF = sample(c(TRUE, FALSE), N, replace = TRUE, prob = c(0.15, 0.85))
)

head(df)
#>   patient_id exam_id age nn_predicted_age    AF
#> 1         1 4787395  27      50.38434 FALSE
#> 2         2 3263944  30      61.17839 FALSE
#> 3         2 1981044  54      52.15138  TRUE
#> 4         2 4232992  68      75.89813 FALSE
#> 5         3 3548531  83      69.45013 FALSE
#> 6         4 3524725  28      62.87170 FALSE

```

3.1 Keep One Exam per Patient

Because some patients have multiple exams, we demonstrate both supported methods:

- "latest": keep the largest exam_id per patient
- "random": randomly select one exam per patient (reproducible by seed)

```

df_latest <- filter_to_unique_patient(df, method = "latest")
df_random <- filter_to_unique_patient(df, method = "random", seed = 99)

nrow(df_latest)
#> [1] 200
length(unique(df$patient_id))
#> [1] 200

```

3.2 Compute Age Gap

```

df_latest <- compute_age_gap(df_latest)
summary(df_latest$age_gap)
#>   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#> -65.774 -20.110  -2.606  -1.337  17.762  55.030

```

3.3 Run Permutation Test

We form two samples: - Group 1: AF == TRUE - Group 2: AF == FALSE

Then run permutation test with a selectable statistic.

```

x <- df_latest$age_gap[df_latest$AF == TRUE]
y <- df_latest$age_gap[df_latest$AF == FALSE]

res_mean <- run_permutation_test(x, y, stat = "mean", B = 500, seed = 123)
res_median <- run_permutation_test(x, y, stat = "median", B = 500, seed = 123)
res_ks <- run_permutation_test(x, y, stat = "ks", B = 500, seed = 123)

res_mean$p_value
#> [1] 0.1457086
res_median$p_value
#> [1] 0.3373253
res_ks$p_value
#> [1] 0.2894212

```

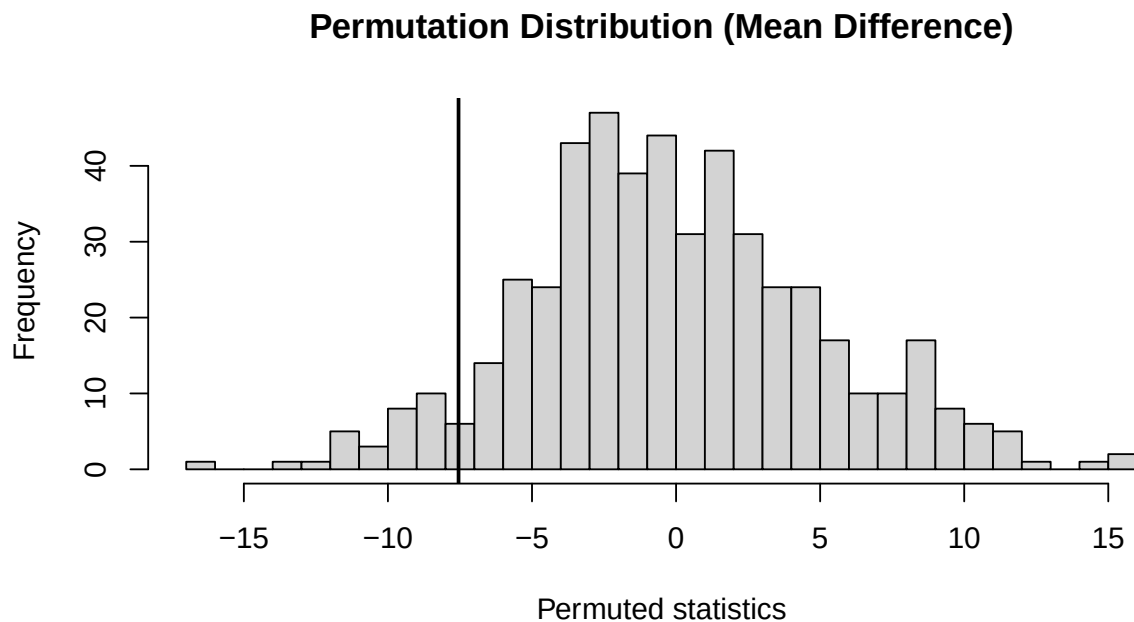
4. Graphs: permutation distribution

A required output is the permutation distribution plot with a vertical line at the observed statistic.

```

hist(res_mean$t_perm, breaks = 30,
     main = "Permutation Distribution (Mean Difference)",
     xlab = "Permuted statistics")
abline(v = res_mean$t_obs, lwd = 2)

```



5. Brief Assumption / Caveat Check

Permutation testing assumes that under H_0 , the group labels are exchangeable.

A practical issue in this dataset is repeated exams per patient, which violates strict independence.

Our default approach is to keep only one exam per patient using `filter_to_unique_patient()`. (An optional extension in the full project is patient-level block permutation.)

6. Testing (testthat) — Sample Outputs Shown

This checkpoint requires writing tests with **testthat** and showing some outputs in the vignette. Below is a short snippet of what our tests validate:

- `compute_age_gap()` correctly creates
$$\text{age_gap} = \text{nn_predicted_age} - \text{age}.$$
- `filter_to_unique_patient()` returns one row per patient (latest method)
- `filter_to_unique_patient(..., method = "random")` is reproducible with the same seed
- `run_permutation_test()` returns the correct structure and valid p-values
- An invalid `stat` argument triggers an error

```
if (requireNamespace("testthat", quietly = TRUE)) {
  testthat::test_file("test_code.R", reporter = "summary")
} else {
  cat("Package 'testthat' is not installed in this environment.\n",
      "Install it to run and display the full test output from test_code.R.\n")
}
#> Warning: package 'testthat' was built under R version 4.5.2
#> code: .....
#>
#> == DONE ==
```

7. References (BibTeX)

```
@dataset{ribeiro_2021_code15,
  author    = {Ribeiro, A. H. and Ribeiro, M. H. and Paixao, G. M. M. and others},
  title     = {CODE-15\%: A Large-Scale Electrocardiogram Dataset for Deep Learning},
  year      = {2021},
  publisher = {Zenodo},
  doi       = {10.5281/zenodo.4916206},
  url       = {https://doi.org/10.5281/zenodo.4916206}
}
```

Ribeiro, A. H., M. H. Ribeiro, G. M. M. Paixao, et al. 2021. "CODE-15%: A Large-Scale Electrocardiogram Dataset for Deep Learning." Zenodo. <https://doi.org/10.5281/zenodo.4916206>.